



Telecom Academy



Module 2: Basic Switch and End Device Configuration

Module 10: Basic Router Configuration

Telecom Academy - Cristian Deacu

Introduction to Networks v7.0
(ITN)



Module Objectives

Module Title: Basic Switch and End Device Configuration

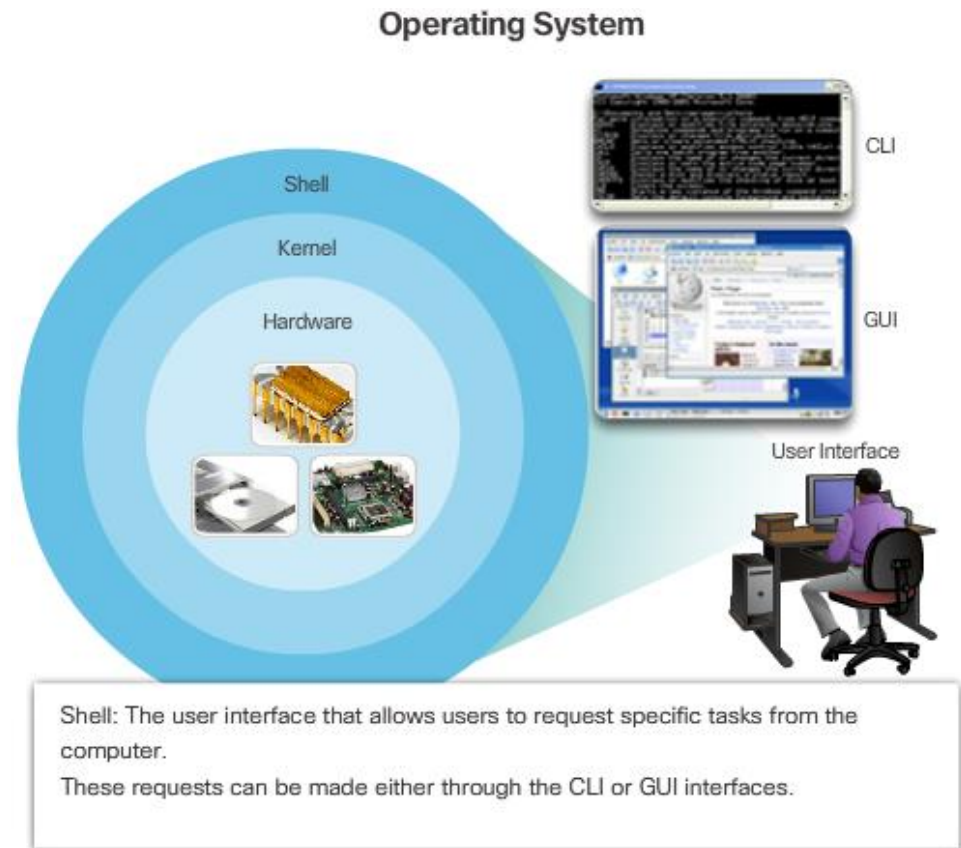
Module Objective: Implement initial settings including passwords, IP addressing, and default gateway parameters on a network switch and end devices.

Topic Title	Topic Objective
Cisco IOS Access	Explain how to access a Cisco IOS device for configuration purposes.
IOS Navigation	Explain how to navigate Cisco IOS to configure network devices.
The Command Structure	Describe the command structure of Cisco IOS software.
Basic Device Configuration	Configure a Cisco IOS device using CLI.
Save Configurations	Use IOS commands to save the running configuration.
Ports and Addresses	Explain how devices communicate across network media.
Configure IP Addressing	Configure a host device with an IP address.
Verify Connectivity	Verify connectivity between two end devices.

2.1 Cisco IOS Access

Cisco IOS Access Operating Systems

- **Shell** - CLI or GUI
- **Kernel**
- **Hardware**



Cisco IOS Access GUI

- A GUI allows the user to interact with the system using an environment of graphical icons, menus, and windows.
- A GUI is more user-friendly and requires less knowledge of the underlying command structure that controls the system.
- Examples of these are: Windows, macOS, Linux KDE, Apple iOS and Android.
- GUIs can fail, crash, or simply not operate as specified. For these reasons, network devices are typically accessed through a CLI.



Purpose of an OS

PC operating system enables a user to do the following:

- Use a mouse to make selections and run programs
- Enter text and text-based commands



CLI-based network operating system enables a network technician to do the following:

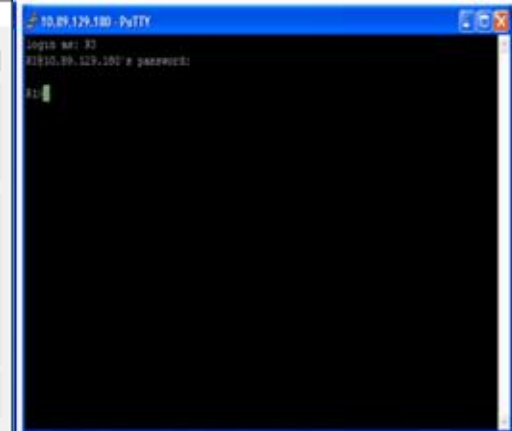
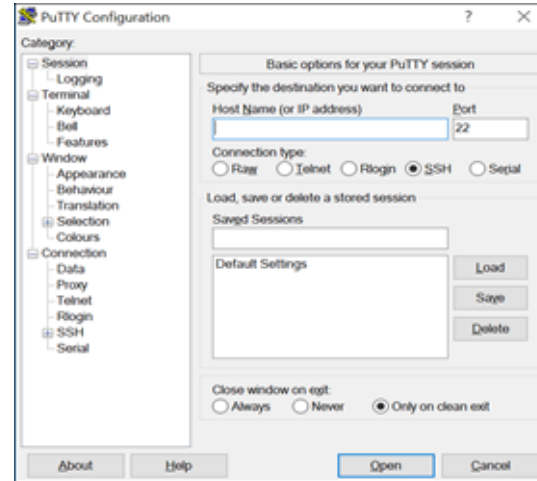
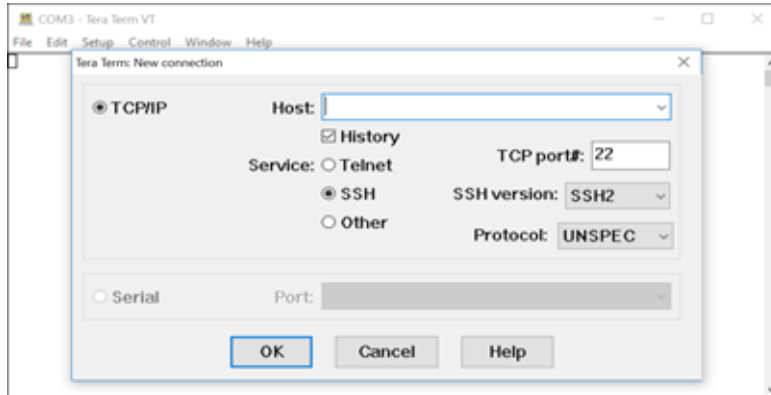
- Use a keyboard to run CLI-based network programs
- Use a keyboard to enter text and text-based commands
- View output on a monitor

```
analyst@secOps ~]$ ls
Desktop Downloads lab.support.files second_drive
[analyst@secOps ~]$
```

Cisco IOS Access

Access Methods

- **Console:**
Rollover (DB9 – RJ45) // USB to USB
 - 1,5m – 2m
 - initial configuration // LAB Scenarios
- Remote login: Telnet, SSH via TCP/IP network
 - everywhere
 - after initial configuration



2.2 IOS Navigation

Primary Command Modes

User EXEC Mode:

- Allows access to only a limited number of basic monitoring commands
- Identified by the CLI prompt that ends with the > symbol

```
Router>
```

```
Switch>
```

Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#
```

```
Switch#
```

Configuration Mode and Subconfiguration Modes

Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config) #
```

Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line) #
```

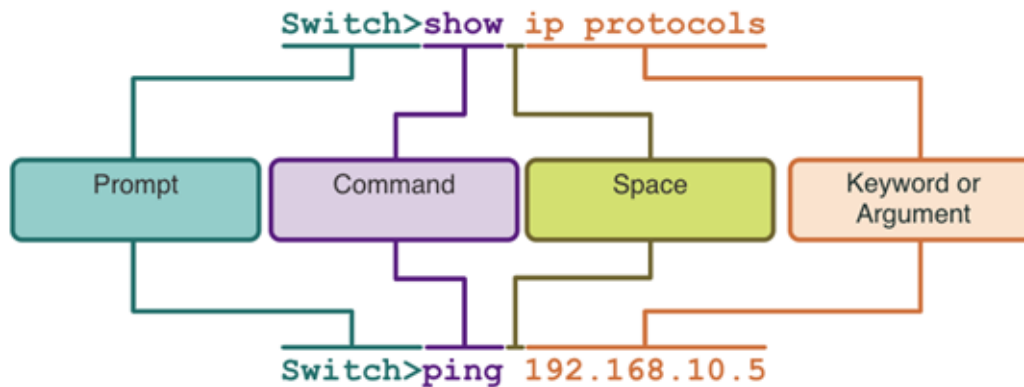
Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if) #
```

2.3 The Command Structure

Basic IOS Command Structure



- **Keyword** – This is a specific parameter defined in the operating system (in the figure, **ip protocols**).
- **Argument** - This is not predefined; it is a value or variable defined by the user (in the figure, **192.168.10.5**).

Basic IOS Command Structure

User EXEC Command - Router>

ping
show (limited)
enable

Privileged EXEC Commands - Router#

all User EXEC commands
debug commands
reload
configure

Global Configuration Commands - Router(config)#

hostname
enable secret
ip route

interface ethernet
serial
dsl

router rip
ospf
eigrp

line vty
console

Interface Commands - Router(config-if)#

ip address
ipv6 address
encapsulation
shutdown/no shutdown

Routing Engine Commands - Router(config-router)#

network
version
auto summary

Line Commands - Router(config-line)#

password
login
modem commands

The Command Structure

IOS Help Features

The IOS has two forms of help available: context-sensitive help and command syntax check.

- Context-sensitive help enables you to quickly find answers to these questions:
 - Which commands are available in each command mode?
 - Which commands start with specific characters or group of characters?
 - Which arguments and keywords are available to particular commands?
- Command syntax check verifies that a valid command was entered by the user.
 - If the interpreter cannot understand the command being entered, it will provide feedback describing what is wrong with the command.

```
Router#ping ?
WORD  Ping destination address or hostname
ip     IP echo
ipv6   IPv6 echo
```

```
Switch#interface fastEthernet 0/1
      ^
% Invalid input detected at '^' marker.
```

Hot Keys and Shortcuts

- The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.
- Commands and keywords can be shortened to the minimum number of characters that identify a unique selection. For example, the **configure** command can be shortened to **conf** because **configure** is the only command that begins with **conf**.

```
Router#con
% Ambiguous command: "con"
Router#con?
configure  connect
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

Hot Keys and Shortcuts (Cont.)

- The table below is a brief list of keystrokes to enhance command line editing.

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.

Hot Keys and Shortcuts (Cont.)

- When a command output produces more text than can be displayed in a terminal window, the IOS will display a “**--More--**” prompt. The table below describes the keystrokes that can be used when this prompt is displayed.
- The table below lists commands that can be used to exit out of an operation.

Keystroke	Description
Enter Key	Displays the next line.
Space Bar	Displays the next screen.
Any other key	Ends the display string, returning to privileged EXEC mode.

Keystroke	Description
Ctrl-C	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl-Z	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl-Shift-6	All-purpose break sequence used to abort DNS lookups, traceroutes, pings, etc.

Note: To see more hot keys and shortcuts refer to 2.3.5.

The Command Structure

Video – Hot Keys and Shortcuts

This video will cover the following:

- Tab key (tab completion)
- Command shortening
- Up and down arrow key
- CTRL + C
- CTRL + Z
- CTRL + Shift + 6
- CTRL + R

```
Switch#>clock set  
% Incomplete command.  
Switch#clock set 19:50:00  
% Incomplete command.
```

The IOS returns a help message indicating that required keywords or arguments were left off the end of the command.

```
Switch#c  
% Ambiguous command: 'c'
```

The IOS returns a help message to indicate that there were not enough characters entered for the command interpreter to recognize the command.

```
Switch#clock set 19:50:00 25 6  
                        ^  
% Invalid input detected at '^'  
marker.
```

The IOS returns a "^" to indicate where the command interpreter can not decipher the command.

2.4 Basic Device Configuration

Device Names

- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a letter
 - Contain no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be less than 64 characters in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Configure Passwords (Cont.)

Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password *password*** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Basic Device Configuration

Encrypt Passwords

- The startup-config and running-config files display most passwords in plaintext.
- To encrypt all plaintext passwords, use the **service password-encryption** global config command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

- Use the **show running-config** command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# show running-config
!
!
line con 0
password 7 094F471A1A0A
login
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
end
```

Basic Device Configuration

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command.

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```


2.5 Save Configurations

Save Configurations

Configuration Files

- There are two system files that store the device configuration:
 - **startup-config** - This is the saved configuration file that is stored in NVRAM.
 - **running-config** - This is stored in Random Access Memory (RAM).
 - **copy running-config startup-config**

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

Save Configurations

Alter the Running Configurations

```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

2.6 Ports and Addresses

Ports and Addresses

Interfaces and Ports



Copper



Fiber-optics



Wireless



2.7 Configure IP Addressing

Manual IP Address Configuration for End Devices

- **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window. T

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address: [. . .]

Subnet mask: [. . .]

Default gateway: [. . .]

☒ Obtain DNS server address automatically

☐ Use the following DNS server addresses:

Preferred DNS server: [. . .]

Alternate DNS server: [. . .]

☐ Validate settings upon exit

Advanced...

OK Cancel

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: [192 . 168 . 1 . 10]

Subnet mask: [255 . 255 . 255 . 0]

Default gateway: [192 . 168 . 1 . 1]

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: [. . .]

Alternate DNS server: [. . .]

☐ Validate settings upon exit

Advanced...

OK Cancel

Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: [2001:db8:acad:10::10]

Subnet prefix length: [64]

Default gateway: [fe80::1]

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: []

Alternate DNS server: []

☐ Validate settings upon exit

Advanced...

OK Cancel

Note: IPv6 addressing and configuration options are similar to IPv4.

Switch Virtual Interface Configuration

To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask command*.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```


10.1 Configure Initial Router Settings

Configure Initial Router Settings

Basic Router Configuration Steps

- Configure the device name.
- Secure privileged EXEC mode.
- Secure user EXEC mode.
- Secure remote Telnet / SSH access.
- Encrypt all plaintext passwords.
- Provide legal notification and save the configuration.

```
Router(config)# hostname hostname
```

```
Router(config)# enable secret password
```

```
Router(config)# line console 0  
Router(config-line)# password password  
Router(config-line)# login
```

```
Router(config)# line vty 0 4  
Router(config-line)# password password  
Router(config-line)# login  
Router(config-line)# transport input {ssh | telnet}
```

```
Router(config)# service password encryption
```

```
Router(config)# banner motd # message #  
Router(config)# end  
Router# copy running-config startup-config
```

Configure Initial Router Settings

Basic Router Configuration Example

- Commands for basic router configuration on R1.
- Configuration is saved to NVRAM.

```
R1(config)# hostname R1
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)# exit
R1(config)# service password encryption
R1(config)# banner motd #
Enter TEXT message. End with a new line and the #
*****
WARNING: Unauthorized access is prohibited!
*****
R1(config)# exit
R1# copy running-config startup-config
```

10.2 Configure Interfaces

Configure Router Interfaces

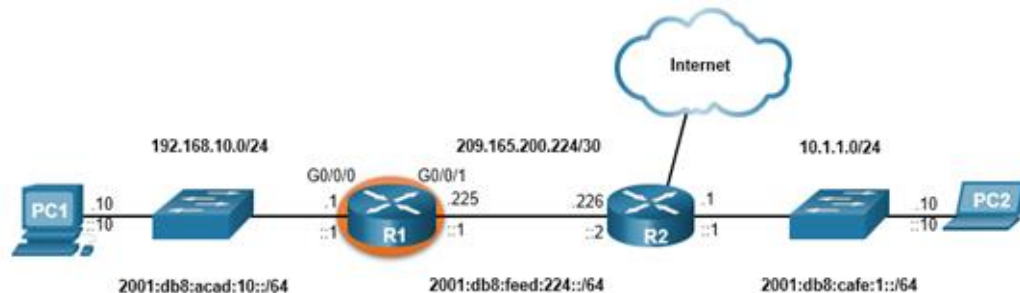
Configuring a router interface includes issuing the following commands:

```
Router(config)# interface type-and-number  
Router(config-if)# description description-text  
Router(config-if)# ip address ipv4-address subnet-mask  
Router(config-if)# ipv6 address ipv6-address/prefix-length  
Router(config-if)# no shutdown
```

- It is a good practice to use the **description** command to add information about the network connected to the interface.
- The **no shutdown** command activates the interface.

Configure Router Interfaces Example

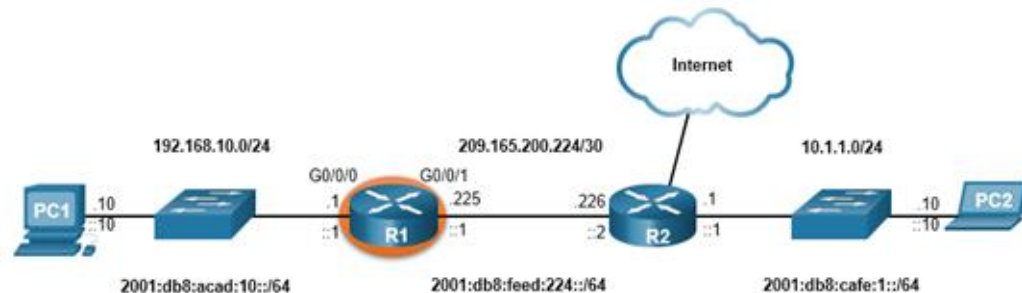
The commands to configure interface G0/0/0 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug  1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug  1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug  1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to up
```

Configure Router Interfaces Example (Cont.)

The commands to configure interface G0/0/1 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Aug 1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Aug 1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1,
changed state to up
```

Verify Interface Configuration

To verify interface configuration use the **show ip interface brief** and **show ipv6 interface brief** commands shown here:

```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0/0     192.168.10.1    YES manual up            up
GigabitEthernet0/0/1     209.165.200.225 YES manual up            up
Vlan1                    unassigned      YES unset  administratively down down
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0      [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1      [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1                     [administratively down/down]
unassigned
R1#
```


Configure Verification Commands

The table summarizes show commands used to verify interface configuration.

Commands	Description
<code>show ip interface brief</code> <code>show ipv6 interface brief</code>	Displays all interfaces, their IP addresses, and their current status.
<code>show ip route</code> <code>show ipv6 route</code>	Displays the contents of the IP routing tables stored in RAM.
<code>show interfaces</code>	Displays statistics for all interfaces on the device. Only displays the IPv4 addressing information.
<code>show ip interfaces</code>	Displays the IPv4 statistics for all interfaces on a router.
<code>show ipv6 interfaces</code>	Displays the IPv6 statistics for all interfaces on a router.

Configure Verification Commands (Cont.)

View status of all interfaces with the **show ip interface brief** and **show ipv6 interface brief** commands, shown here:

```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0/0     192.168.10.1    YES manual up            up
GigabitEthernet0/0/1     209.165.200.225 YES manual up            up
Vlan1                    unassigned      YES unset  administratively down down
R1#
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0     [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1     [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1                    [administratively down/down]
unassigned
R1#
```

Configure Verification Commands (Cont.)

Display the contents of the IP routing tables with the **show ip route** and **show ipv6 route** commands as shown here:

```
R1# show ip route
< output omitted>
Gateway of last resort is not set
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L       209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

```
R1# show ipv6 route
<output omitted>
C   2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C   2001:DB8:FEED:224::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:FEED:224::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#
```

Configure Verification Commands (Cont.)

Display statistics for all interfaces with the **show interfaces** command, as shown here:

```
R1# show interfaces gig0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)
  Description: Link to LAN
  Internet address is 192.168.10.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:01, output 00:00:35, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output      drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    1180 packets input, 109486 bytes, 0 no buffer
    Received 84 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles

<output omitted>

R1#
```

Configure Verification Commands (Cont.)

Display IPv4 statistics for router interfaces with the **show ip interface** command, as shown here:

```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled
```

<output omitted>

R1#

Configure Verification Commands (Cont.)

Display IPv6 statistics for router interfaces with the **show ipv6 interface** command shown here:

```
R1# show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is
FE80::868A:8DFF:FE44:49B0
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:10::1, subnet is 2001:DB8:ACAD:10::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF44:49B0
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds

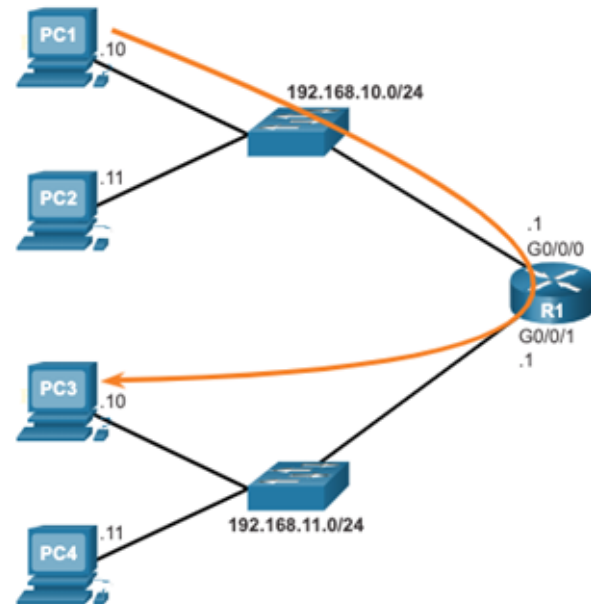
R1#
```

10.3 Configure the Default Gateway

Configure the Default Gateway

Default Gateway on a Host

- The default gateway is used when a host sends a packet to a device on another network.
- The default gateway address is generally the router interface address attached to the local network of the host.
- To reach PC3, PC1 addresses a packet with the IPv4 address of PC3, but forwards the packet to its default gateway, the G0/0/0 interface of R1.



Note: The IP address of the host and the router interface must be in the same network.

Configure the Default Gateway

Default Gateway on a Switch

- A switch must have a default gateway address configured to remotely manage the switch from another network.
- To configure an IPv4 default gateway on a switch, use the **ip default-gateway ip-address** global configuration command.



